

Development of A Cross-Platform Application Blue-Collar Marketplace with Prescriptive Analytics Using Linear Programming Algorithm

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Abstract

Unemployment remains a persistent issue in the Philippines, especially among low- and middle-income earners who lack access to immediate and flexible work opportunities. To address this, a cross-platform application was developed to facilitate job matching for blue-collar workers. The developed system integrates prescriptive analytics through a linear programming algorithm to provide optimized and fair pricing suggestions for job posts, helping employers make data-informed decisions while ensuring equitable compensation for freelancers. Data inputs for the algorithm were sourced from the Public Employment Service Office (PESO) and Katherine Townhomes. Additional features such as identity verification and a feedback system were incorporated to promote trust and service accountability. Developed using the Rapid Application Development (RAD) model, the project used an iterative approach, comprising planning, design, construction, and deployment phases, to ensure user-centered outcomes. The application was evaluated using the ISO/IEC 25010 software quality model, with a focus on functional suitability, performance efficiency, usability, and reliability. Evaluation results revealed high user satisfaction and system effectiveness, with the platform receiving a rating of 4.98 out of 6 from a diverse group of IT professionals and regular users. The developed system not only delivers a functional and accessible job marketplace but also contributes to bridging employment gaps by leveraging technology to foster fair labor practices, transparency, and economic empowerment. The project demonstrates the potential of intelligent platforms to address employment challenges in developing countries by creating inclusive, data-driven, and user-centric digital solutions.

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CCS Concepts

• Information systems; • Information systems applications; • Decision support systems applications; • Data analytics;

Keywords

applied computing, operations research, linear programming, software and its engineering, rapid application, development, blue-collar jobs, job marketplace, prescriptive analytics, cross-platform application development, linear programming

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1 INTRODUCTION

Employment remains a complex and persistent challenge in the Philippines, particularly for low-income individuals and those without access to quality education or professional training. Despite the country's economic growth in recent years, a substantial portion of the population continues to face barriers to securing stable and fulfilling employment. These barriers include limited access to formal job opportunities, underemployment, and a lack of inclusive systems accommodating individuals with varying educational backgrounds and skill levels [1]. Simultaneously, rapid urbanization and demanding professional work have left many high-income earners with limited time for managing day-to-day personal or household tasks. This growing disparity between labor supply and demand highlights an urgent need to reassess how work is distributed, accessed, and valued across various sectors of society. Understanding employment not merely as a matter of economic necessity but as a pathway to social equity highlights the importance of inclusive, flexible, and technology-driven solutions. In the Philippine context, where informal work and digital connectivity intersect,

exploring alternative employment systems is crucial for addressing unemployment and adapting to evolving labor demands.

This paper explores these challenges in depth, drawing from existing labor trends, government initiatives, and socioeconomic data to identify gaps in the current employment landscape. It also lays the foundation for proposing potential innovations to make job opportunities more accessible and responsive to the diverse needs of the Filipino workforce.

1.1 Background of the Study

Unemployment remains a pressing socioeconomic concern, particularly in developing countries such as the Philippines. It refers to individuals who are actively seeking employment but are unable to secure suitable work. Globally, an estimated 192 million people faced this challenge in 2022 due to limited access to education, scarce job opportunities, and inadequate training [2]. The World Bank reported that the global unemployment rate in 2023 stood at 5.1%, representing approximately 178 million individuals struggling to find decent work opportunities [3]. The consequences of unemployment extend beyond personal economic hardship, as they also threaten social stability and long-term national development.

One alarming outcome of persistent unemployment is its correlation with increased crime rates. Socioeconomic stressors, such as income inequality, limited opportunities, and unemployment, are often cited as key factors that drive individuals toward unlawful activities in an attempt to meet their basic needs [4, 6]. These actions not only affect direct victims but also severely impact the families of offenders, especially children, by compromising their health, emotional well-being, education, and future economic mobility [14]. Addressing unemployment is thus critical not only for economic progress but also for promoting public safety and social equity [6].

In the Philippines, as of April 2023, approximately 2.04 million Filipinos were unemployed, a figure that, while seemingly modest in percentage terms, affects millions of households [1]. To address this, several programs have been introduced. The Public Employment Service Office (PESO) facilitates access to job vacancies, particularly for blue-collar workers, and connects employers with potential hires [9]. Similarly, JobStart Philippines, under the Department of Labor and Employment (DOLE), provides life skills training to improve youth employability [9]. The Government Internship Program (GIP) also offers individuals aged 18–30 the opportunity to gain work experience and develop practical competencies [13].

While these government-led initiatives show promise, their implementation faces several limitations. Many programs lack long-term support structures, provide minimal stipends, and rely heavily on traditional employer-employee arrangements that may not align with the realities of low-income individuals. For many, daily subsistence takes precedence over participating in extended training or internship programs. Consequently, short-term, hourly-based employment, often associated with blue-collar or gig work, emerges as a more practical and immediate alternative. However, access to such jobs remains constrained due to trust issues, informal hiring practices, and the prevalence of recruitment scams [7, 12].

Recognizing these challenges, this study proposes a technology-driven employment platform focused primarily on blue-collar job

opportunities, aligning with the United Nations Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth, which advocates for inclusive and productive employment for all [5]. This cross-platform system, accessible via mobile and web, aims to empower both freelancers and employers by providing secure job postings, optimized pricing, and identity verification mechanisms, thereby enhancing trust and safety in the hiring process. By expanding access to verified, short-term employment opportunities for underserved populations, the platform aims to enhance economic resilience, reduce informality in the labor market, and promote inclusive economic participation [10, 11].

By adopting this approach, it contributes to broader efforts to promote sustainable economic growth, reduce inequalities in employment access, and build a more responsive and secure labor market, particularly for low-income and underrepresented workers in the Philippines.

The study aims to design and evaluate a cross-platform job-matching system that supports efficient interaction between freelancers and employers. Specifically, it seeks to implement an administrative dashboard for effective monitoring and visualization of key operational data, integrate descriptive and prescriptive analytics through a Linear Programming (LP) model to support pricing recommendations, and assess the overall quality and performance of the developed system using the ISO/IEC 25010 software quality model, with emphasis on functional suitability, performance efficiency, usability, and reliability.

1.2 Significance of the Study

Providing accessible job opportunities for low- to middle-income earners is a shared vision among many policymakers seeking to ensure a quality and sustainable standard of living for citizens. The developed system may contribute to this goal by promoting fair, transparent, and technology-driven employment practices that foster safer and more sustainable ways to earn a living.

The primary beneficiaries of this initiative are individuals who struggle to find short-term or flexible work opportunities that align with their skills and daily availability. Through this platform, freelancers can easily browse available jobs suited to their abilities and preferences, enabling them to earn immediate income through short-duration tasks. This flexibility allows workers to manage their schedules more effectively compared to traditional eight-hour work arrangements, ultimately supporting their financial needs and reducing work-related stress.

For employers, particularly those from higher-income households, the developed system provides a convenient means of hiring workers on demand for occasional or unplanned tasks such as household cleaning, delivery, or personal assistance. This arrangement promotes efficiency and cost savings, as employers pay only for completed tasks rather than maintaining long-term employment contracts.

Additionally, the developed system enhances safety and accountability through features such as user identity verification, requiring government-issued identification before registration. This verification process fosters trust between employers and freelancers while ensuring a secure digital environment. By providing legitimate earning opportunities, the platform also contributes to reducing

petty crimes motivated by financial hardship, thereby supporting community safety and social stability.

At the policy level, the project aligns with United Nations Sustainable Development Goal 8: Decent Work and Economic Growth, by fostering inclusive employment and reducing unemployment within local communities. Beyond its social impact, the project also offers value to researchers and organizations by generating business intelligence and actionable insights through data analytics. These findings can inform future innovations, enhance labor market strategies, and inform the development of data-driven policies that promote equitable economic participation.

The developed system holds both societal and research significance, empowering workers, aiding employers, promoting ethical employment practices, and contributing to national and global goals of sustainable and inclusive economic growth.

2 REVIEW OF RELATED LITERATURE

This section presents relevant studies, government initiatives, and digital innovations that address unemployment and promote livelihood development. It also includes discussions on conceptual and architectural frameworks supporting the developed system. The literature review highlights the intersection of technology, data analytics, and socioeconomic policies in providing employment opportunities and enhancing national productivity.

In response to persistent unemployment, government agencies and policymakers have implemented programs to provide accessible job opportunities and skills training. In the Philippines, the Public Employment Service Office (PESO) serves as a local hub connecting employers and job seekers. It provides listings for various positions such as service crew, production workers, and call center agents, bridging the gap between workforce supply and labor demand [9].

Another initiative, JobStart Philippines, implemented by the Department of Labor and Employment (DOLE), aims to enhance employability among the youth by offering training and life skills development [9]. After completing the training phase, participants are referred to partner employers for internships, giving them practical experience and professional exposure. Similarly, the Government Internship Program (GIP) provides short-term work opportunities in public offices to individuals aged 18–30, enabling them to gain valuable workplace experience while receiving financial assistance [13]. These initiatives collectively promote workforce readiness and contribute to economic resilience.

However, despite their benefits, such programs face challenges in scalability, funding, and accessibility. Many participants from low-income backgrounds prioritize immediate income over long-term training, which limits their participation in programs. Consequently, flexible, short-term employment remains an attractive and necessary option for many workers.

The rise of digital technologies has given birth to various job-matching platforms that address unemployment by linking workers to short-term or task-based opportunities. One notable example is TaskRabbit, a U.S.-based platform that connects freelancers with clients seeking assistance with services such as plumbing, delivery, and home repairs [12]. Similarly, Handy focuses on professional home services like cleaning, furniture assembly, and general repairs,

emphasizing service quality and user satisfaction through verified professional workers and 24/7 support [12].

Another example, Snagajob, streamlines recruitment for part-time and full-time service jobs in the hospitality industry by allowing job seekers to maintain digital profiles that serve as employment applications. This approach streamlines hiring and minimizes paperwork, making the process more efficient for both employers and job seekers [12].

These existing platforms demonstrate how technology can enhance employment access, improve user experience, and foster trust through identity verification and performance-based feedback systems, features that also inform the design of the proposed platform in this study.

The integration of data analytics into employment systems has gained traction in both private and public sectors. Research by Galdon-Sanchez and Gil [10] found that the adoption of big data and analytics provides actionable insights for improving productivity, innovation, and job creation, particularly among small and medium-sized enterprises (SMEs). Data-driven decision-making enables organizations to develop strategies that more closely align with labor market trends, ultimately reducing unemployment and enhancing competitiveness.

During the COVID-19 pandemic, predictive analytics played a crucial role in tracking unemployment patterns. Simionescu and Cifuentes-Faura [15] leveraged Google Trends data to estimate monthly unemployment rates in Spain and Portugal, providing policymakers with insights to anticipate economic challenges and take timely action. This example underscores how analytics tools can support governments and organizations in managing workforce-related issues, a concept that informs the prescriptive analytics aspect of the research project.

The relationship between employment and economic growth has been widely studied in development economics. Haider et al. [11] emphasized that countries with higher employment intensity exhibit stronger GDP growth, reinforcing the need to create jobs that contribute meaningfully to national output. Similarly, Correa [8] found that unemployment hurts GDP, while moderate inflation may stimulate economic activity by encouraging consumption and investment. These findings highlight the importance of balanced macroeconomic policies that support job creation while maintaining economic stability.

Furthermore, Sustained productivity growth, supported by technological investment and infrastructure development, can contribute to the creation of high-value jobs and overall economic advancement. Thus, leveraging digital innovation to connect workers with employment opportunities not only addresses individual livelihood concerns but also contributes to broader national development goals.

The reviewed literature demonstrates that both government programs and digital job-matching platforms play significant roles in addressing unemployment. However, traditional government programs often lack flexibility, while existing commercial platforms are primarily targeted toward developed economies. Bridging these gaps requires a localized, inclusive, and data-driven solution tailored to the Philippine context.

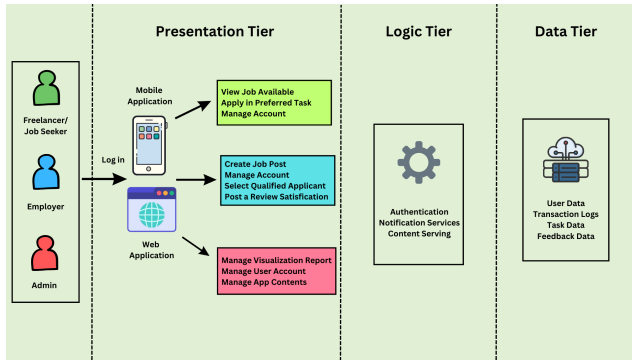


Figure 1: Three-tier Architecture

The research project responds to this need by integrating prescriptive analytics and secure identity verification into a cross-platform employment marketplace. Through these innovations, it aims to enhance job accessibility, improve employer-employee trust, and contribute to sustainable economic growth aligned with United Nations Sustainable Development Goal 8: Decent Work and Economic Growth [5].

The system employs a three-tier architecture, comprising the presentation, logic, and data layers, each serving a distinct role in ensuring efficiency and scalability. Figure 1 shows the system's architectural framework and how these layers interact to support its overall functionality. The presentation layer, developed using PHP and JavaScript, provides an intuitive user interface for employers, freelancers, and administrators. It emphasizes responsive design and seamless accessibility across both desktop and mobile platforms to enhance user experience. The Logic Layer, implemented primarily with Python scripts, serves as the system's core processing component. It executes pricing recommendations based on the Linear Programming model while managing the business logic responsible for job matching and analytics generation. Finally, the Data Layer utilizes a MySQL database to store essential records, including user profiles, job postings, applications, and transaction histories. This layer ensures efficient data retrieval, consistency, and integrity across all system operations, thereby supporting the platform's stability and reliability.

3 RESEARCH METHODOLOGY

3.1 Requirements Planning Phase

The development process began with the Requirements Planning phase, where the research team analyzed existing digital job platforms and conducted informal interviews with potential users to identify functional requirements, pricing challenges, and usability concerns. Based on these findings, the system was conceptualized as a digital job marketplace for blue-collar workers and employers in Metro Manila, focusing on short-term services such as cleaning, delivery, handyman work, and personal assistance.

During this phase, core system functionalities were identified, including distinct user roles (freelancer, employer, and administrator), job posting and matching workflows, and analytics-driven decision

support tools. Particular emphasis was placed on addressing inconsistent job pricing and the lack of data-driven recommendations in existing platforms, which led to the inclusion of a prescriptive analytics component using Linear Programming.

3.2 User Design Phase

In the User Design phase, interface prototypes were developed using tools such as Figma and WordPress, later transitioning to a PHP-based architecture for full front-end and back-end integration. Iterative design sessions were conducted, incorporating continuous feedback from potential users to refine navigation, usability, and visual layout.

An administrative dashboard was designed to visualize job market trends, service category distributions, and pricing analytics. During this phase, the analytics framework was finalized, combining descriptive analytics (for historical and trend analysis) and prescriptive analytics (for optimized job pricing recommendations).

3.3 Construction Phase

The Construction phase involved full-scale system development. The core system architecture was implemented using PHP, while analytics computations were handled through Python libraries. Major system functionalities such as user authentication, job posting, **application matching, messaging, and dashboard reporting were developed and tested.**

3.3.1 Linear Programming Model Specification. As part of the prescriptive analytics component, a Linear Programming (LP) model was developed to assist employers in determining fair, competitive, and market-aligned job pricing.

- Model Type:** The model is a single-objective Linear Programming model. No normalization or scaling was required, as all monetary values were expressed in consistent currency units (PHP).
- Decision Variable:** The decision variable is the recommended job price, expressed in Philippine Pesos (xxx).
- Objective Function:** The objective of the model is to maximize employer value by setting a competitive price that balances affordability and freelancer attractiveness. Formally, the objective is expressed as Maximize $Z = x$ {Maximize } $Z = x$ Maximize $Z = x$, reflecting the goal of achieving the highest feasible price within market and system constraints, thereby improving job acceptance rates while maintaining employer budget efficiency.
- Constraints:** The pricing model is subject to a minimum market price constraint ($x \geq P_{\min}$ ge $P_{\min}$) to ensure compliance with acceptable market or freelancer rates, a maximum employer budget constraint ($x \leq P_{\max}$ le $P_{\max}$) to prevent exceeding the employer's stated budget, and a demand-based constraint ($x \leq P_{\text{demand}}$ le P_{demand}) to maintain competitiveness under prevailing market demand. A non-negativity constraint ($x \geq 0$ ge 0) is also imposed.
- Parameter Definitions.** Solution Method: The model parameters are defined in Table 1. The LP model was solved using the Simplex Method, producing feasible and optimal pricing recommendations. These recommended prices were

Table 1: Parameter Definitions

Parameter	Description	Unit
P_{min}	Minimum acceptable market price	PHP
P_{max}	Employer's maximum budget	PHP
P_{demand}	Market demand-based price ceiling	PHP

integrated into the system dashboard and presented to employers during job posting.

3.4 Cutover Phase

Following system construction, the Cutover phase focused on deployment, integration testing, and system refinement. Integration testing ensured compatibility across all modules, while usability testing validated feature completeness and navigation efficiency. Identified issues were addressed through iterative improvements to enhance reliability and user experience.

3.5 System Evaluation

The system was evaluated based on the ISO/IEC 25010 software quality standards, with particular focus on functional suitability, performance efficiency, usability, and reliability. Functional suitability assessed whether the system features met the specified requirements, while performance efficiency measured responsiveness, processing speed, and stability. Usability was evaluated in terms of ease of use, learnability, and interface intuitiveness, and reliability was examined the system's stability under varying usage conditions. Evaluation was conducted through a combination of technical testing and user feedback from IT professionals and general users, providing a comprehensive measure of both system quality and user satisfaction.

4 RESULTS AND DISCUSSIONS

This section presents the organized documentation and analysis of the system's development outcomes. The primary objective was to design and implement a cross-platform job marketplace that connects blue-collar freelancers with employers seeking short-term labor assistance. The discussion covers the system's functionalities, structural design, and evaluation results based on user testing and software quality assessment.

The platform serves as a digital employment marketplace that bridges the gap between blue-collar workers and employers seeking temporary or task-based services. The platform facilitates quick job matching for various work categories, including household cleaning, personal assistance, delivery, handyman tasks, and yard maintenance.

The development of the platform initially began with a front-end prototype using WordPress, before transitioning to a PHP-based architecture to achieve full integration between the front-end and back-end components. The final system provides a seamless user experience through the following core functionalities:

- Job posting and application management: Employers can create and manage job listings, while freelancers can apply for available tasks.

- Data visualization dashboard: Administrators can view analytics on job market trends, including service category distribution and seasonal demand.
- Prescriptive pricing analytics: A Linear Programming algorithm generates optimized and fair price suggestions for job postings.
- Messaging system: Facilitates communication between freelancers and employers for coordination and feedback.
- Review and rating system: Employers can evaluate freelancers' performance to promote accountability and reliability.

These features collectively contribute to a functional, user-oriented system that addresses the challenges of informal hiring and limited job accessibility in the blue-collar sector.

Figure 2 shows the administrator dashboard, which serves as the control center for system monitoring. The dashboard displays key metrics such as the total number of freelancers, employers, and job postings. Additionally, it includes data visualization charts depicting the distribution of service categories and seasonal job trends. The administrator is also provided with tables for pricing optimization and management of predefined pricing parameters, supporting informed decision-making for job valuation.

4.1 Summary of Software Evaluation Results

Table 2 presents the evaluation results based on the ISO/IEC 25010 software quality standard, showing an overall mean rating of 4.98, interpreted as *Agree*. This reflects a high level of user satisfaction across the evaluated quality characteristics. The findings indicate that the system meets the essential standards of functional suitability, usability, reliability, security, portability, and performance efficiency.

Overall, feedback from both respondent groups demonstrates a consistently positive perception of the platform, confirming that the system is functional, secure, user-friendly, and efficient in accordance with the ISO/IEC 25010 evaluation framework.

5 CONCLUSION AND RECOMMENDATION

5.1 Conclusion

The developed system effectively achieves its primary objective of creating an innovative, user-centered job marketplace tailored for blue-collar workers and employers in the Philippines. It facilitates efficient and inclusive connections, addressing the growing demand for short-term labor in Metro Manila. A key innovation is the integration of prescriptive analytics through a Linear Programming algorithm, which generates dynamic, data-driven pricing recommendations, promoting fairness, transparency, and equitable compensation. The platform also features a dynamic administrative dashboard that provides visual insights into user activity, job trends, service categories, and financial summaries, supporting informed decision-making and operational accountability. Evaluation under the ISO/IEC 25010 framework demonstrated high functional suitability, performance efficiency, usability, and reliability, with both IT professionals and general users confirming the system's effectiveness, security, and ease of use. Overall, the system delivers a functional, data-driven, and socially responsive digital employment solution, enhancing inclusive employment opportunities and

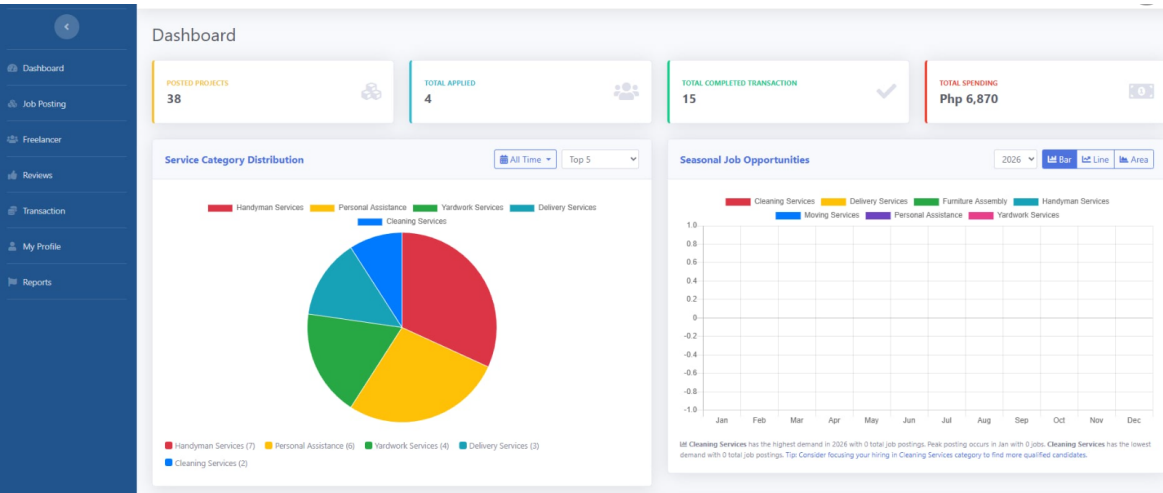


Figure 2: Admin Dashboard Page

Table 2: Summary Results on ISO/IEC-25010

Respondents	Criteria	Results	Interpretation	Average Score	Interpretation
Regular Users (e.g., Freelancer, Employer)	Functionality	5.44%	Strongly Agree	4.83%	Agree
	Usability	5.21%	Agree		
IT Professionals	Performance	5.19%	Strongly Agree	5.13%	Agree
	Usability	5.15%	Agree		
	Reliability	5.19%	Strongly Agree		
	Security	5.02%	Agree		
	Portability	5.15%	Agree		
	Functionality	5.09%	Agree		
Average Result				4.98%	Agree

contributing to the sustainable development of the gig economy in the Philippines.

5.2 Recommendations

Future work should focus on enhancing the system through the implementation of concrete, feature-driven improvements that increase usability, transparency, and decision accuracy. Key priorities include the development of an automated job classification and tagging mechanism to standardize service categories, reduce manual input, and improve job discoverability. Additionally, the introduction of structured review scoring and reputation metrics, based on clearly defined performance criteria such as reliability, communication, and service quality, would enhance trust and accountability within the platform. Lastly, the incorporation of rule-based decision support features, including pricing guidance and job-matching recommendations derived from historical platform data and predefined thresholds, would further support informed decision-making for both freelancers and employers.

6 DECLARATION OF ARTIFICIAL INTELLIGENCE RESEARCH ASSISTANCE

The authors used AI-assisted writing tools to improve the clarity and readability of portions of the manuscript. These tools did not contribute to the research design, data analysis, or conclusions. The authors reviewed all content and take full responsibility for the final manuscript.

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